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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/151,277	01/18/2021	Zack W. BLACKMON	24000NV-000667-US	9784

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EXAMINER
NGUYEN, THANG H

ART UNIT	PAPER NUMBER
2831	

NOTIFICATION DATE	DELIVERY MODE
10/11/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/151,277	Applicant(s) BLACKMON et al.	
	Examiner THANG H NGUYEN	Art Unit 2831	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 7/24/23.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1 and 4-22 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,4-6,9-13 and 15-21 is/are rejected.

8) ☑ Claim(s) 7-8,14 and 22 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/18/21 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 1; 4-5, 12-13, 15, and 17-18 as best understood and is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Hepworth et al. (US 2020/0390149).

As per claim 1: Hepworth et al. discloses a capsule 4 for an aerosol-generating device 1, comprising: a housing 60 defining inlet openings, outlet openings (see fig. 1; wherein the both end of the capsule/cartridge are having openings wherein element 64 is located), and a chamber 66 between the inlet openings and the outlet openings, the chamber 66 having a longest dimension extending from at least one of the inlet openings to a corresponding one of the outlet openings (as shown in fig. 1); an aerosol-forming substrate 63 within the chamber 66 of the housing 60; and a heater 70 extending into the housing 60, the heater 70 including a first end section, an intermediate section, and a second end section, the intermediate section being disposed within the aerosol-forming substrate 63 in the chamber 66 (see fig. 1; wherein the 1st end section is adjacent to the inlet openings 68, and the 2nd section is adjacent to the outlet openings 68 (opposite of the other openings 68), and the intermediate section is located in the middle of the housing), wherein the housing 60 has a longitudinal axis (as shown in fig. 1,

wherein the capsule 4 is insert into the article), and the longest dimension of the chamber 66 extends along the longitudinal axis of the housing 60, and wherein the housing 60 includes a body portion 62 and a base portion 64, the body portion 62 including a proximal end and a distal end, the base portion 64 configured to engage with the distal end of the body portion 62 (see attached fig. 3).

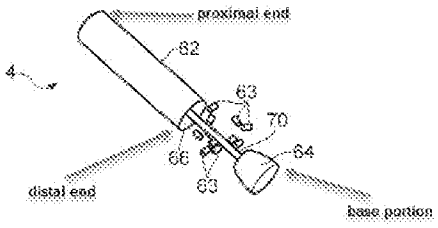


FIG. 3

As per claims 4-5, 12-13, 15, 17: Hepworth et al. discloses the capsule 4 for the aerosol-generating device 1, wherein the base portion 64 of the housing 60 defines an air channel in fluidic communication with the chamber 66 via the inlet openings 68 (as shown in fig. 1, Para. [0031]); and wherein the body portion 62 of the housing 60 defines the inlet openings, the outlet openings, and the chamber 66 (see fig. 1; wherein the both end of the capsule/cartridge are having openings wherein element 64 is located); and wherein the aerosol-forming substrate includes a plant material; and wherein the plant material includes tobacco (see Para. [0031-0032]); and wherein the intermediate section of the heater is between the inlet openings and the outlet openings (as shown in fig. 1, wherein the middle portion of the heater is located in between the inlet and outlet opening); and wherein at least one annular member 94 engaged with the housing 60 (as shown in fig. 5E).

As per claim 18: Hepworth et al. discloses an aerosol generating device 1, comprising: a capsule 4 including a housing 60 including a body portion 62 and a base portion 64, the body portion 62 including a proximal end and a distal end, the base portion 64 configured to engage with the distal end of the body portion 62 (as shown in attached fig. 3 above), an aerosol-forming substrate 63, and a heater 70, the housing 60 defining inlet openings, outlet openings (see fig. 1; wherein the both end of the capsule/cartridge are having openings wherein element 64 is located, and a chamber 66 between the inlet openings and the outlet openings (as shown in fig. 1); a mouthpiece 50 configured to engage with

the capsule 4 so as to be in fluidic communication with the chamber 66 via the outlet openings (see Para. [0023-0024]); and a device body 12 configured to receive and retain the capsule 4 and the mouthpiece 50, the device body 12 including a power source 26 configured to supply an electric current to the heater 70 to heat the aerosol-forming substrate 63 (as shown in fig. 2).

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claim(s) 6, 9-11, 16, 21 as best understood and is/are rejected under 35 U.S.C. 103 as being unpatentable over Hepworth et al. (US 2020/0390149).

As per claims 6, 9-10, and 21: Hepworth et al. discloses the capsule 4 for the aerosol-generating device 1, and the body portion 62 of the housing 60. However, Hepworth et al. does not explicitly disclose wherein the body portion of the housing includes a first body component and a second body component; and wherein the first body component and the second body component are configured to clamp onto the base portion; and wherein the first body component and the second body component are identical and complementary structures.

On the other hand, Hepworth et al. discloses the body portion 62 of the housing 60 is one integral piece with a cavity/chamber within. However, a person having ordinary skill in the art would know that having the body portion of the housing includes a first body component and a second body component; and wherein the first body component and the second body component are configured to clamp onto the base portion; and wherein the first body component and the second body component are identical and

complementary structures can only deal with to make separable, because such modification still would not change the function of the capsule/cartridge or the aerosol generating device after all but to further enhance the functionality of the body portion to adapt to other components within as the manufacture desire structure. Since, it has been held that constructing a formerly integral structure in various elements involves only routine skill in the art. *Nerwin v. Erlicnrnan,* 168 USPQ 177, 179.

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for the aerosol generating device of Hepworth et al. by having the body portion of the housing includes a first body component and a second body component; and wherein the first body component and the second body component are configured to clamp onto the base portion; and wherein the first body component and the second body component are identical and complementary structures as taught by the instant invention to further enhance the functionality of the body portion to adapt to other components within as the manufacture desire structure.

As per claim 11: Hepworth et al. discloses the capsule 4 for the aerosol-generating device 1, the heater 70 and the aerosol forming substrate 63. However, Hepworth et al. does not explicitly disclose wherein the aerosol-forming substrate includes a first consolidated form and a second consolidated form disposed on opposite sides of the intermediate section of the heater.

On the other hand, Hepworth et al. mentioned that the heater can be in different forms/shapes within the capsule (see figs. 8A-C, Para. [0049-0052]). However, a person having ordinary skill in the art would know that having different consolidated forms disposed on opposite sides of the heater can only deal with to make separable because such modification still would not change the function of the capsule/cartridge in the aerosol generating device after all but to further enhance more options to the consumer to have different flavors/materials instead only one flavor/material within the aerosol generating device. Since, it has been held that constructing a formerly integral structure in various elements involves only routine skill in the art. *Nerwin v. Erlicnrnan,* 168 USPQ 177, 179.

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for the aerosol generating device of

Hepworth et al. by having the aerosol-forming substrate includes a first consolidated form and a second consolidated form disposed on opposite sides of the intermediate section of the heater as taught by the instant invention to further enhance more options to the consumer to have different flavors/materials instead only one flavor/material within the aerosol generating device. Further, having multiple consolidated forms within a capsule/cartridge is well known and common knowledge in the art of electronic cigarette.

As per claim 16: Hepworth et al. discloses the capsule 4 for the aerosol-generating device 1, and the heater having a generally planar form (see Para. [0049]). However, Hepworth et al. does not explicitly disclose wherein the intermediate section of the heater has a winding form.

On the other hand, Hepworth et al. discloses the heater can be in different forms and shapes (as shown in figs. 8A-C). However, a person having ordinary skill in the art would know that having the heater has a winding form can only deal with change in shape because such modification still would not change the function of the heater or the capsule/cartridge after all but to further enhance the effective heating, stable and reliable heating to the aerosol substrate within the capsule/cartridge. A change in shape is generally recognizing as being within the level of ordinary skill in the art. In re Dailey, 149 USPQ 47 (CCPA 1976).

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the capsule for the aerosol generating device of Hepworth et al. by having the intermediate section of the heater has a winding form as taught by the instant invention to further enhance the effective heating, stable and reliable heating to the aerosol substrate within the capsule/cartridge.

Claim(s) 19-20 as best understood and is/are rejected under 35 U.S.C. 103 as being unpatentable over Hepworth et al. (US 2020/0390149) in view of Reevell (US 2018/0214645).

As per claims 19-20: Hepworth et al. discloses the aerosol generating device 1, the capsule 4, the device body 12 and the mouthpiece 50. However, Hepworth et al. does not disclose wherein the device body includes a door configured to open to permit an insertion of the capsule and the mouthpiece and

configured to close to retain the capsule and the mouthpiece; and wherein the mouthpiece includes a mouth end and an opposing capsule end, the capsule end being larger than the mouth end and configured to prevent a disengagement of the mouthpiece from the capsule when the door of the device body is closed.

Reevell discloses the aerosol-generating device, wherein the device body 1 includes a door 20 configured to open to permit an insertion of the capsule (31/32/33) and the mouthpiece (21/22/23) and configured to close to retain the capsule (31/32/33) and the mouthpiece (21/22/23) (as shown in figs. 1-2, 4). Although, neither Reevell nor Hepworth et al. discloses exactly the structure as recited above. However, a person having ordinary skill in the art would know that it would have been an obvious matter of design choice, because such modification still would not change the function of the capsule as well as the mouthpiece and the aerosol generating device as recited above. Further, since applicant has not disclosed the special structure above would solve any stated problem or is for any particular purpose and it appear that the invention would perform equally well with the prior art of record.

Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claim invention to further modify the aerosol generating device of Hepworth et al. having the device body includes a door configured to open to permit an insertion of the capsule and the mouthpiece and configured to close to retain the capsule and the mouthpiece as taught by Reevell and the mouthpiece includes a mouth end and an opposing capsule end, the capsule end being larger than the mouth end and configured to prevent a disengagement of the mouthpiece from the capsule when the door of the device body is closed as taught by the instant invention to further enhance the convenience of changing/replacing of the capsule/cartridge after all.

Allowable Subject Matter

Claims 7-8, 14, and 22 are objected to as being dependent upon a rejected base claim.

Response to Arguments

Applicant's arguments with respect to claim(s) 1, and 18 have been considered but the new ground of rejection does rely on a reference applied in the prior rejection of record because Applicant did not argue or specifically challenged in the argument regard to the reference.

Conclusion

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to THANG H NGUYEN whose telephone number is (571)270-0288. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Abdullah Riyami can be reached on 571-270-3119. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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